

Fractional Derivatives in Biomathematical Models with Memory: A Critical Discussion

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Abstract

In recent years, the application of fractional derivatives in mathematical models has gained significant popularity. In the case of time-fractional derivatives, one of the main reasons for their use lies in their nonlocal property, which can overcome the limitations of ordinary differential equation models, which are purely local and might fail to describe memory-dependent processes. The most common approach, often called fractionalisation, is based on the direct replacement of the classical derivatives in the ODE models, with fractional ones. When they are compared to real data, fractionalised models are often shown to provide better fitting results. The most common interpretation of this is that fractionalised models keep track of the history, while local models do not. However, while the physical meaning of a classical derivative is clear, the same cannot be said for fractional derivatives. Therefore, the relationship between modelling assumptions and mathematical equations remains unclear. Here, we introduce and critically discuss the fractionalisation approach by considering two representative examples of fractionalised biomathematical models. In our discussion, we address several properties of fractional operators that may impose significant limitations in their applications. However, the key question on which we would like to reflect is: *is a fractionalised model still a model?*

Keywords: fractional models; fractionalisation; fractional derivatives;
Riemann-Liouville derivative; Caputo derivative;

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1 Introduction

Biomathematics relies on the translation of biological questions and assumptions into a mathematical language. The main advantage is that, through a mathematical description, it might be easier to identify possible answers and get new biological intuitions. In particular, when approaching to a biological problem, the role of the modeller is constantly challenged by the need of finding the right balance between description accuracy and mathematical tractability. In other words, when modelling, the translation to mathematics needs to be, at the same time, both accurate and easily readable. This is generally not an easy task, since the world we are living in is surely not simple, hence, from a certain point of view, the main job of a biomathematician is to transform complexity into simplicity. In very broad terms, if one wants to address a biological question, the addition of extra mathematical complexity to mathematical models is a good idea, as long as it adds novel and relevant information. As an example, let us consider one of the most classic modelling problems in biomathematics, which studies the growth of a population. If $N(t)$ represents the size of a population at time t , the ordinary differential equation (ODE) $N'(t) = aN(t)$, with $a \in \mathbb{R}$, describes its evolution. The crucial assumption behind this model is that a population grows or decays, proportionally to its size. Such assumption might work well in describing, for example, an initial phase of bacterial growth (see e.g. (Maier et al., 2009, Chapter 3)), but it is not true in general, since no population can grow exponentially at all times. The refinement of the biological assumptions often leads to extra mathematical complexity, which might be worth to deal with, if one wants to have a more accurate model. For instance, the logistic equation $N'(t) = aN(1 - N/K)$ models the population growth limited by a maximal carrying capacity $K > 0$ (see e.g. (Murray, 2002, Chapter 1)). More generally, the goal is to find, given a set of biological assumptions, a suitable function f that describes the variation rate of the population, i.e. $N' = f$. Such f can be very general, and it might depend on t and N , as well as on other parameters or variables. In some cases, the temporal complexity of a phenomenon might need more advanced tools.

Notable examples of more advanced tools are the *fractional derivatives*. In the last couple of decades, considerable effort has been devoted to incorporating these operators into mathematical models. Throughout this paper, we will discuss their use and, often, misuse, with a focus on time-dependent models with functions of the single variable $t \geq 0$. Historically, the idea of derivatives of non-integer order dates back to the late 17th century, originating from correspondence between Leibniz and L'Hôpital. Since then, mathematicians have defined a wide variety of fractional derivatives (see e.g. de Oliveira and Tenreiro Machado (2014)), which share the property of being integro-differential operators, extending the concept of differentiability to a real, complex or variable differentiation order. For a more comprehensive overview, the reader is referred to the classical textbooks on fractional calculus by Diethelm (2010); Kilbas et al. (2006); Miller and Ross (1993); Podlubny (1999). Among the most widely known and used fractional derivatives are the Riemann-Liouville and Caputo definitions. The first one is derived from the corresponding fractional integral, which is obtained by extending the Cauchy formula for repeated integration to non-integer

orders. The application of a generalised form of the Fundamental Theorem of Calculus then yields the Riemann-Liouville derivative. In turn, the Caputo derivative is defined by reversing the order of differentiation and fractional integration. When the fractional order is $\alpha \in (0, 1]$, these are defined, respectively, as

$${}^{RL}D_0^\alpha u(t) := \frac{d}{dt} \int_0^t k_\alpha(t, s) u(s) ds \quad \text{and} \quad {}^CD_0^\alpha u(t) := \int_0^t k_\alpha(t, s) u'(s) ds,$$

with $k_\alpha(t, s) := \frac{(t-s)^{-\alpha}}{\Gamma(1-\alpha)}$. Alternative kernel choices for $k_\alpha(t, s)$ might define other fractional derivatives and fractional operators. Nevertheless, our point of discussion aims at being very general, and purely focused on how fractional operators are incorporated into models, rather than on the differences between their different definitions. Therefore, in the following, with $D_0^\alpha u$ we might indicate a fractional operator, without necessarily specifying its type. Within our discussion, it makes sense to include also more recent operators, such as the Caputo-Fabrizio and Atangana-Baleanu-Caputo operators, which are defined by a nonsingular kernel. However, as a side note, one should keep in mind that these last two operators share some common modelling criticisms related to restrictions on the choice of the initial conditions, as described by [Diethelm et al. \(2020\)](#).

There exist important differences between classical and fractional derivatives. For instance, the Riemann-Liouville derivative of a constant $C \in \mathbb{R}$ is a classical example, since it is not zero, but a singular function in 0, as ${}^{RL}D_0^\alpha C = Ct^{-\alpha}/\Gamma(1-\alpha)$ for $\alpha < 1$. Here, our main point of interest is that, for $\alpha \in (0, 1)$, all of the mentioned fractional derivatives and operators are nonlocal, since the calculation of their value at time t requires to know the function $u(s)$ at all previous time $s \in [0, t)$. From a physical point of view, nonlocality supports the interpretation that fractional operators share the property of keeping a sort of memory. In these terms, the kernel $k_\alpha(t, s)$ can be seen as a weight function for the entire history of u up to time t . This argument is commonly used as a motivation for the application of fractional derivatives in modelling certain memory-dependent processes. But how can we use fractional derivatives in modelling a phenomenon, for instance the population growth? In addressing this question, we should keep in mind that the modeller's main interest is to get a mathematical model able to describe, explain or predict some biological insight.

In this manuscript we discuss the most common approach, in the literature, for including time-fractional derivatives in mathematical models, which relies on the *ad-hoc* replacement of the classical derivatives with fractional ones. For this reason, [Dokoumetzidis et al. \(2010a\)](#) call it *fractionalisation*. In order to question the validity of this approach, we select two examples of fractionalised models from the literature and present them in Sections 3 and 4. In the final section, we further discuss our results in light of the applicability of fractional derivatives in modelling.

2 Classical models and their fractionalisation

The fractionalisation method generally comes at the end of a classical modelling process. The initial modelling process is generally based on the application of first

principles, such as, for instance, the Law of Mass Action (see, e.g., the classical textbook by [Murray \(2002\)](#)). Here, the modelling assumptions are translated into ordinary differential equations, which describe the variation rate of certain quantities. The fractionalisation process corresponds to the mathematical generalisation of such a system, where the generalisation refers to the differentiation order α , which passes from being fixed to 1 to vary within the interval $(0, 1]$. For instance, our general model of population growth $N' = f$ in its fractionalised version reads as $D_0^\alpha N = f$. This approach provides a mathematical generalisation, as $D_0^1 N = N'$, however, a mathematical generalisation does not necessarily correspond to a model generalisation. Can the fractional differential equation $D_0^\alpha N = f$ still be considered as a population model? Certain authors have noticed that fractional generalisations are not consistent with respect to dimensional units, therefore, an extra dimensional balancing parameter is often placed on the right-hand side. In fact, dimensional inconsistency is just the tip of an iceberg representing a bigger problem: in our population model equation, the function f is specific to a set of assumptions for the variation rate of the population size N . By just replacing N' with a fractional derivative, we are immediately losing the connection between left and right hand side of the equation. More importantly, we are replacing something which has a clear meaning, with something that we are no longer able to either understand or interpret with comparable clarity. For completeness, we recall that, already [Dokoumetzidis et al. \(2010a\)](#) highlight the fact that dimensional inconsistency is the evident indicator of a bigger problem related to the fractionalisation. However, their criticism only concerns multi-order fractional systems. They show that, if one considers a mass-conserving ODE system, by replacing the classical derivative in the i -th equation with a fractional one of order $\alpha_i \in (0, 1]$, the mass conservation property is inevitably lost, unless all α_i are equals.

A completely different approach relies on the modification of the right hand side of the equation, as we discussed above, regarding the identification of a suitable function f in the population model. This approach is especially used in data-driven modelling when certain kinetic functions are unknown, for instance if one wants to identify certain dynamics that do not satisfy the law of mass action. Interestingly, in a series of papers, finally summarised in ([Angstmann et al., 2021](#)), the authors derive, from a stochastic process, different fractional differential equation models where a Riemann-Liouville derivative appears, as a result of some modelling assumptions, on the right-hand side of the equation, while on the left-hand side is still present the classical derivative. We briefly describe their approach in [Box 1](#).

Given their nature, the inclusion of fractional derivatives adds extra complexity to a model, and here comes the question of why we should use such tools. In the following sections, we consider two different ODE models and their fractionalised versions, which have been proposed in the literature as improvements to the original ones. These representative examples have been selected, among a set of recent well-cited works, especially because they provide all the details permitting comparison with our results. However, we expect that similar conclusions can be drawn from most of the fractionalised models present in the literature. Hence, we will not enter the biological details of the following examples, but discuss their general modelling approach. Indeed, in the following two examples, by comparing the mathematical solutions with real

data, the authors claim that fractional systems might be more suitable and more efficient in describing certain processes, namely chickenpox epidemics and human liver activity. Nevertheless, in our work, we show how it is possible to obtain even better results through naive modifications of the original ODE systems, without the need of using fractional derivatives. In particular, we show that, if one wants to evaluate the goodness of a model only based on data fitting, simpler and more interpretable systems might still perform better than fractionalised ones. More importantly, there might be no need to sacrifice model interpretability to gain accuracy in data representation.

3 A fractionalised MSEIR model applied to chickenpox data

The following ODE system

$$\begin{cases} m'(t) = b(e + i + r) - \delta m, \\ e'(t) = \beta i(1 - m - e - i - r) - \epsilon e - be, \\ i'(t) = \epsilon e - (\gamma + b)i, \\ r'(t) = \gamma i - br, \end{cases} \quad (1)$$

represents a reduced version of the MSEIR model, which has been used to model the transmission of diseases such as measles, rubella or mumps ([Hethcote, 2000](#)). Here, m , e , i and r represent, respectively, the fractions of individuals born with temporary immunity, individuals exposed to the disease, infected individuals, and recovered ones. Note that, the fraction of susceptible individuals s is not explicitly considered with an equation, as $s = 1 - m - e - i - r$. The other parameters b , δ , ϵ , γ are positive constants that define the transition rates across the compartments. Three different fractionalised versions of it were studied by [Qureshi and Yusuf \(2019\)](#), where they replace the classical derivatives in System (1) with fractional operators and raise the system parameters to the power α to ensure dimensional consistency. In their work, the authors test the performance of three different fractional operators, namely the Caputo, Caputo-Fabrizio (CF) and Atangana-Baleanu-Caputo (ABC) operators, and, by fitting the system to chickenpox data, they compare the fractionalised system with the original ODE counterpart. By minimising the Euclidean distance between the solution $i(t)$ and the infected cases from data, they estimate the parameter β and the fractional order $\alpha \in [0, 1]$. Since in the classical case the order α is fixed to 1, better performances of the fractional systems should not surprise. The results, reported in [Figure 1](#), obviously show that all the three operators perform better than the classical derivative, and the use of the ABC operator leads to the best prediction. It remains to understand if such improvement is solely due to the estimation of the extra parameter α , or also to the inclusion of memory effects. In other terms, can we say that the classical model performs worse because it is purely local? Are fractional operators really needed to improve the model results? In order to do so, we consider

the following system

$$\begin{cases} m'(t) = b(e + i + r) - \delta m, \\ e'(t) = \beta i(1 - m - e - i - r) - \epsilon e^\alpha - be, \\ i'(t) = \epsilon e^\alpha - (\gamma + b)i, \\ r'(t) = \gamma i - br, \end{cases} \quad (2)$$

which constitutes another mathematical generalisation of the original system, where the linear term ϵe is replaced by ϵe^α , with $\alpha > 0$. Power law kinetics of this type have been studied theoretically and empirically, see e.g. [Schnell and Turner \(2004\)](#) and [Taghvaei et al. \(2020\)](#). With this generalisation, by comparing the solution $i(t)$ to the data and estimating the optimal parameters α and β , we are able to halve the error of the ABC generalisation, which corresponds to the best result by [Qureshi and Yusuf \(2019\)](#), as reported in the legend of Figure 1. While the comparison with the data apparently provides a strong support to our nonlinear modification, without further investigation on the data used and on the modelling assumptions, this is surely not enough to conclude that we built a better model. On the other hand, the fact that the nonlinear system (2) performs much better than the fractionalised ones does not mean that chickenpox infection dynamics are not affected by memory effects. Nevertheless, in this case, the nonlinear system (2) might still be seen as a model, where the transition rate from the compartment E to I does not respect the Law of Mass Action, as it is sub-linear. On the other hand, it results extremely difficult to retrieve the biological assumptions behind any of the three fractional cases considered by [Qureshi and Yusuf \(2019\)](#), since the fractionalised systems do no longer describe the variation rate of the 4 compartments. Lastly, the poor fitting of the original MSEIR model to the data needs specific investigations, but knowing the assumptions, which are clear in this model, helps to identify if some of them do not apply to chickenpox transmission in this particular situation or if some other crucial assumptions have been neglected.

Before moving to the next example, it is important to highlight another modelling challenge typical of the fractionalisation approach: how to set appropriate initial conditions. For simplicity, consider a more general epidemiological model that describes the spreading of an infection since its very first appearance and suppose we have some data in a certain time interval. While determining the origins of an epidemic is an extremely challenging task, this does not constitute a problem for classical ODE models: since they are local, any available data point can serve as an initial condition. In this case, the past and unknown history will not affect the future dynamics. In contrast, since fractionalised models are nonlocal systems, the initial conditions should be imposed very accurately: in this case, it is necessary to know where, in time, we should set the initial condition. This difference with respect to the ODE models is generally not considered during fractionalisation. For instance, with reference to Figure 1, the chickenpox epidemic does not actually begin with 628 cases of infection at $t = 0$, but much earlier, see e.g. the original dataset ([Tang et al., 2017](#)). In such a case, the initial conditions should be defined over an interval $[0, t_0]$, with 628 cases reported at a certain time $t_0 > 0$, which needs to be identified. Imposing them at time

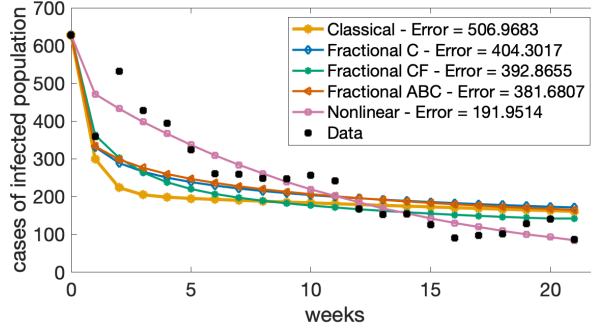


Fig. 1 A comparison of the classical (1), fractionalised and nonlinear system (2) with epidemiological data. In the nonlinear case (2), $\alpha = 0.7935$ and $\beta = 2.5937$. In the classical case (1), $\beta = 2.7897$. In the three fractional cases, $\alpha = 0.54061$ and $\beta = 4.2515$ for Caputo, $\alpha = 0.55127$ and $\beta = 5.24$ for Caputo-Fabrizio, $\alpha = 0.55127$ and $\beta = 2.14$ for Atangana-Baleanu-Caputo. For each case, the error reported in the plot legend is the Euclidean distance between the infected cases and the rescaled model solution $Ni(t)$, where $N := I_0/i_0$, with I_0 denoting the first data point and i_0 the initial condition.

$t = 0$ neglects all the previous history and, implicitly, contradicts the assumption that memory effects should be included. This issue has been previously discussed in the literature, see, for example, [Sabatier and Farges \(2021\)](#), where the authors provide simple examples demonstrating that setting proper initial conditions is non-trivial in the context of fractional differential models.

4 A fractionalised model of the human liver activity

The following ODE system

$$\begin{cases} Z'(t) = -aZ(t) + bW(t) \\ W'(t) = aZ(t) - (b + d)W(t), \end{cases} \quad (3)$$

with $a, b, d > 0$, was originally proposed by [Čelechovská \(2004\)](#) for modelling human liver functions. In particular, she studied how Bromsulphthalein (BSP), which is a substance injected into the blood stream, is taken up by the liver. The variables $Z(t)$ and $W(t)$ represent, respectively, the amount of BSP in the blood and in the liver. The rates of transfer a , b and d were originally estimated from the data, which are available for both components Z and W . [Baleanu et al. \(2020\)](#) modify the above system by replacing the classical derivatives with the Caputo-Fabrizio operators of order α , as follows

$$\begin{cases} {}^{CF}D^\alpha Z(t) = -a^\alpha Z(t) + b^\alpha W(t) \\ {}^{CF}D^\alpha W(t) = a^\alpha Z(t) - (b^\alpha + d^\alpha)W(t), \end{cases} \quad (4)$$

where the original parameters are raised to the power α in order to maintain dimensional consistency. By fitting the numerical solutions to the experimental data, via the minimisation of the L^∞ -norm of their difference, the best order α for the Caputo-Fabrizio operator is $\alpha = 0.9869$. Since ${}^{CF}D^1 Z(t) = Z'(t)$, and ${}^{CF}D^\alpha Z(t)$ is continuous with respect to $\alpha \in [0, 1]$ for any $Z \in C^1[0, T]$ (Caputo and Fabrizio, 2015; Losada and Nieto, 2015), it should not surprise that the error does not considerably improve, as it is shown in Table 1.

Similarly to the previous section, as a test, we consider a nonlinear generalisation of the original system

$$\begin{cases} Z'(t) = -aZ(t)^\alpha + bW(t) \\ W'(t) = aZ(t)^\alpha - bW(t) - dW(t). \end{cases} \quad (5)$$

where a is now meant to have units $[Z]^{1-\alpha}/[t]$. By minimising the sum of the two absolute errors e_Z and e_W , that represent the L^∞ -norm of the difference between the data and the solution, respectively, Z and W , we obtain a better overall fitting with respect to the previously considered cases, as reported in Table 1. However, while the fitting of W has substantially improved, the fitting of the Z component got worst, especially with respect to the fractional case (4) by Baleanu et al. (2020). Yet, for our case (5), there exists a range of values of α such that e_Z and e_W are both smaller than the corresponding ones in the fractional case. When restricting the minimisation to such a range, the overall improvement is relatively small, however the nonlinear model still performs better, see last row in Table 1. Therefore, if data fitting was the only criterion for evaluating a model, then our nonlinear generalisation is surely the best one among the three. Nevertheless, in this example, the estimated parameter α , both in the fractional and in our nonlinear generalisation, remains very close to its original value 1 and the minimised errors remain comparable to the original one. Hence, especially given the relatively small improvements in the other cases, it naturally comes the question if adding fractional complexity is really worth it. Indeed, on the contrary, these results could potentially be interpreted as an additional support to the original model, since both the fractional and nonlinear modifications can be seen as small perturbations of it. We conclude this discussion by remarking that, as in the previous example in Section 3, the values of the parameters a, b and d were fixed to the values originally estimated by Čelechovská (2004) for the original model (3). Their values are therefore strictly interrelated to the ODE system and they have a meaning within that framework. When changing the original model, one might need to consider a completely new parameter estimation.

5 Conclusion

There is a vast scientific literature on the application of fractional time derivatives in mathematical biology, where they are often used to replace classical derivatives in ordinary differential equation models. From a modelling point of view, however, the inclusion of fractional derivatives generally introduces additional complexity in both the derivation and the interpretation of the model. In particular, as discussed in this

Model	α	Absolute error			Relative error		
		e_Z	e_W	$e_Z + e_W$	η_Z	η_W	$\eta_Z + \eta_W$
Original (3)	1	10.3543	12.8474	23.2017	0.0414	0.0866	0.1280
Fractional (4)	0.9869	8.8253	11.6643	20.4896	0.0353	0.0787	0.1140
Nonlinear (5)	1.0241	13.2811	3.7350	17.0161	0.0531	0.0252	0.0783
Nonlinear (5) (α constrained)	1.0042	8.8250	11.2807	20.1057	0.0353	0.0761	0.1114

Table 1 The absolute and relative fitting errors for the original model, compared to the fractional (values from [Baleanu et al. \(2020\)](#)) and to the nonlinear case (5). The distance from the data is measured by the discrete L^∞ -norm of the difference between data $\tilde{Z}_n$ and $\tilde{W}_n$ and the numerical solutions $Z(t_n)$, $W(t_n)$. Here, $e_Z := \min_n \|Z(t_n) - \tilde{Z}_n\|_\infty$ and $e_W := \min_n \|W(t_n) - \tilde{W}_n\|_\infty$, while $\eta_Z := e_Z / \|\tilde{Z}\|_\infty$ and $\eta_W := e_W / \|\tilde{W}\|_\infty$. The values on the third row, corresponding to the nonlinear model (5) are the ones obtained by minimising the sum of the absolute errors e_Z and e_W . On the forth row the minimisation is applied only among the values of α such that e_Z and e_W are both smaller than the corresponding values in the fractional case.

article, the fractionalisation approach may transform a clearly interpretable model into a physically meaningless fractional system.

The nonlocality of fractional derivatives suggests interpreting them as memory operators, and indicates that fractional differential equations may be suitable for the description of non-Markovian dynamics. In many cases, fractionalised models are validated primarily through data fitting, and the results are used to support the general claim that memory effects are important. However, while passing to fractional generalisations we might gain accuracy in fitting the data, we lose clarity in the understanding and interpretation of the model. In this article, we argue that, in many situations, *less is more*, especially when dealing with the fractionalisation of the temporal derivatives in ordinary differential equation models. Modelling is not just data fitting: although achieving good agreement with data is surely important, the best mathematical model is the one that we can still understand.

It is important to recall that models incorporating memory have a long-standing history, including many classical examples. The well-known epidemiological SIR model was considered by [Kermack and McKendrick \(1927\)](#) as a special case of their more general age-of-infection model. In his seminal studies, [Volterra \(1926\)](#) incorporated historical effects into his predator–prey model through integro-differential equations, accounting for the fact that food intake by predators does not result in immediate population growth. Another classical example is [Hutchinson \(1948\)](#), who introduced a discrete finite delay in the logistic model of population growth. Since then, integro-differential equations and delay differential equations have become widely used and well-established within the mathematical modelling community.

By contrast, the use of fractional derivatives is much more recent and still controversial, especially with regard to the fractionalisation approach. An interesting attempt to incorporate fractional derivatives into compartmental models was described by [Dokoumetzidis et al. \(2010b\)](#). In that work, the authors start by considering a model in the form of an ODE system. Their approach can be described with three steps: (1) the system is written in its corresponding integral form; (2) the integrals are generalised by the inclusion of a Riemann-Liouville kernel; (3) the integral system is finally differentiated once again, leading to a fractional differential system, where a classical derivative appears on the left-hand side and Riemann-Liouville fractional derivatives appear on the right-hand side. While the fractional terms describe

the inter-compartment fluxes, rather than the variation rates, without further assumptions, this approach can be regarded as methodologically equivalent to what we call fractionalisation.

In turn, [Angstmann et al. \(2021\)](#) developed a more consistent modelling approach, which we briefly summarise in Box 1. Their work builds on the connection between the Riemann-Liouville fractional derivative and heavy-tailed waiting time probability distributions, in particular Mittag-Leffler distributions, previously described and applied by [Hilfer and Anton \(1995\)](#), [Scalas et al. \(2000\)](#) and others. Assuming that the waiting time in a compartment is Mittag-Leffler distributed, Angstmann *et al.* derive evolution equations in which the Riemann-Liouville derivative acts on the fluxes between compartments, while the classical derivative describes the usual rate of change of the population inside the compartment. Their approach provides a physical interpretation of the fractional system, where the fractional differentiation order α is a parameter related to the prescribed waiting time distribution.

Nevertheless, modellers might still ask whether fractional derivatives are really necessary for modelling memory-dependent processes, and under which conditions they are more suitable than classical tools such as integro-differential equations or delay differential equations. While these questions are central to the development of robust models, works such as those by [Angstmann et al. \(2021\)](#) are fundamental to understanding the physical meaning of these operators. Indeed, given the current knowledge on fractional derivatives, the derivation of a fractional model from clear modelling assumptions is probably more relevant to the general question of the physical interpretation of fractional derivatives than to the biological questions that motivated the derivation of the model in the first place.

In addition, the mathematical and numerical analysis of fractional differential equations can help in addressing some of the many research questions related to the interpretation of the differential operators. For instance, every fractional derivative is different, and different is the concept of memory related to it. What does memory mean in applications? How does it differ from one fractional derivative to another? In which cases fractional differential equations are more appropriate than differential equations with delay or more general integro-differential equations? All of these issues constitute fundamental questions that are worth considering and are summarised in Box 2.

In conclusion, fractional derivatives may be a powerful tool, but they must be handled with care. Rather than generalising pre-existing ODE models to fractional differential systems, a step-by-step derivation from first principles should be preferred. Angstmann *et al.* suggest one way, but there remains a wide range of open possibilities worth investigating. Quoting George Box, “all models are wrong, but some are useful”, it is fundamental to keep a clear relationship between modelling assumptions and mathematical equations.

Box 1: The Mittag-Leffler function and fractional derivatives

The Mittag-Leffler function. The Mittag-Leffler function was introduced in the early 1900s by the Swedish mathematician Gösta Mittag-Leffler and it is defined by

$$E_\alpha(t) := \sum_{k=0}^{\infty} \frac{t^k}{\Gamma(\alpha k + 1)}, \quad \alpha > 0, t \in \mathbb{R}.$$

It constitutes a generalisation of the exponential function, as $E_1(t) = e^t$. It has been observed that, when $\alpha \in (0, 1)$ and $t \geq 0$, then $E_\alpha(-t^\alpha)$ constitutes a *survival function*, with a power-law asymptotic decay, $E_\alpha(-t^\alpha) \sim t^{-\alpha}/\Gamma(1-\alpha)$. While this function was not introduced in the context of fractional calculus, it has become fundamental in this field. An important example is that $E_\alpha(-\lambda t^\alpha)$ is solution to the Caputo eigenvalue problem $D_{0,t}^\alpha \rho(t) = -\lambda \rho(t)$, $\lambda \in \mathbb{R}$. The interested reader might consult the textbook by [Gorenflo et al. \(2014\)](#), dedicated to this topic.

Mittag-Leffler waiting times and fractional derivatives. Here we briefly summarise the approach proposed by Angstmann *et al.* to incorporate Riemann-Liouville fractional derivatives in compartmental models. For a more detailed explanation see their review paper ([Angstmann et al., 2021](#)). Consider a compartment where particles enter subject to a flux $q(t)$ and remain according to a survival probability $\Phi(t-s)$ that depends on the amount of time since their first arrival. For simplicity let us assume that at time zero there are no particles. Then, at any $t > 0$, the number of particles is given by

$$\rho(t) = \int_0^t q(s) \Phi(t-s) ds.$$

By differentiating, and after some manipulations, we obtain the following integro-differential equation

$$\rho'(t) = q(t) - \int_0^t K(t-s) \rho(s) ds, \quad (6)$$

where $K(t-s)$ is a function called *memory kernel* and it is defined via its Laplace transform $\mathcal{L}[K] := (1 - s\mathcal{L}[\Phi])/\mathcal{L}[\Phi]$. Consider a Mittag-Leffler distribution for the waiting time inside the compartment, i.e. $\Phi(t) = E_\alpha(-(\nu t)^\alpha)$, with $\alpha \in (0, 1)$ and scaling parameter $\nu > 0$. Therefore, since $\mathcal{L}[\Phi] = s^{\alpha-1}(s^\alpha + \nu^\alpha)^{-1}$, it results

$$\int_0^t K(t-s) \rho(s) ds = \mathcal{L}^{-1} \{ \nu^\alpha s^{1-\alpha} \mathcal{L}[\rho] \} = \nu^\alpha D_{0,t}^{1-\alpha} \rho(t),$$

where $D_{0,t}^{1-\alpha} \rho(t)$ is the Riemann-Liouville derivative of order $1-\alpha$ of $\rho(t)$ and the last equality comes from inverting the Laplace transform of the Riemann-Liouville derivative of $\rho(t)$ with $\rho(0) = 0$. Therefore (6) can be written as

$$\rho'(t) = q(t) - \nu^\alpha D_{0,t}^{1-\alpha} \rho(t). \quad (7)$$

As a side note, we remark that, the assumption of the waiting time to be exponentially distributed *i.e.* $\Phi(t) = e^{-\nu t}$, corresponds to the the Law of Mass Action. Indeed, in this case (6) reduces to the classical ODE $\rho'(t) = q(t) - \nu \rho(t)$.

This approach has been mainly applied within the epidemiological context, for instance in SIR models, where the Mittag-Leffler waiting time was assumed to regulate either the infectivity ([Angstmann et al., 2016a](#)), or the recovery ([Angstmann et al., 2016b](#)), or both ([Angstmann et al., 2017](#)). Other fields of application involve pharmacokinetics ([Mtshali and Jacobs, 2023](#); [Byun et al., 2022](#)), urban bike sharing system ([Graef et al., 2019](#)). As a side note, recently, [Angstmann et al. \(2025\)](#) adapt the above procedure to derive delayed SIR models, by using a delay exponential distribution instead of the Mittag-Leffler one.

Box 2: Some fundamental questions in fractional derivative models

Nonlocality

- How does nonlocality relate to memory?
- What type of memory are we including in the model? (for instance, how does Caputo memory differs from Riemann-Liouville?)
- How do we set appropriate initial conditions?

Interpretation

- A fractional derivative in time is not a variation rate with memory. What does it represent?
- How can we interpret the model outcomes from a biological point of view?
- How can we discuss fractional models with experimentalists?

Solutions

- On which function space does the fractional model admits solutions?
- The numerical resolution of a fractional-order model requires more memory and computational cost with respect to classical ODE models.

Dimensional units

- Is the model consistent with respect to dimensional units?

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Declarations

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The data used in this article are available from the references cited in the main text.

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